

# L1: The History of Network Science

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Let's talk now, rather briefly, about the history of network science.

First, it is important to emphasize that the term “*network*” has been used for decades in different disciplines.

For example, computer scientists would use the term to refer exclusively to computer networks, sociologists have been studying social networks for more than 50 years, and of course, mathematicians have been studying graphs for more than two centuries.

So what is new in network science?

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## The Roots of Network Science

Network Science certainly leveraged concepts and methods that were developed earlier in Graph Theory, Statistical Mechanics and Nonlinear Dynamics in Physics, Computer Science algorithms, Statistics and Machine Learning. The list below shows the key topics that each of these disciplines contributed to Network Science.

- **Graph theory:**
  - Study of abstract (*mostly static*) graphs
- **Statistical mechanics:**
  - Percolation, phase transitions
- **Nonlinear dynamics:**
  - Contagion models, threshold phenomena, synchronization
- **Graph algorithms:**
  - Network paths, clustering, centrality metrics
- **Statistics:**
  - Network sampling, network inference
- **Machine learning:**
  - Graph embeddings, node/edge classification, generative models
- **Theory of complex systems:**

- Scaling, emergence

There are ***two main differences*** however between these disciplines and Network Science.

***First***, Network Science focuses on real-world networks and their properties – rather than on regular or random graphs, which are easier to analyze mathematically but not realistic. Most of the earlier work in graph theory or physics was assuming that networks have that kind of simple structure.

***Second***, Network Science provides a general framework to study complex networks independent of the specific application domain. This unified approach revealed that there are major similarities and universal properties in networks, independent of whether they represent social, biological or technological systems.